Atharva Prasad Jakhadi

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Summary

Industrial and Systems Engineering graduate student with experience in simulation, analytics, process improvement and supply chain management. Skilled in modeling complex systems using Simio and FlexSim, with a strong focus on optimizing operations and driving efficiency.

Education

SUNY at Binghamton *Aug 2025 – Present*
Master of Science in Industrial and Systems Engineering | GPA – 3.7/4.0

Vellore Institute of Technology *Aug 2021 – May 2025*
Bachelor of Technology in Biotechnology | GPA – 7.2/10.0

Relevant Coursework

Applied Probability and Statistics, Enterprise Systems Engineering, Operations Management and Supply Chain, Engineering Project Management, Advanced Engineering Analytics, Modeling and Simulation, Quality Assurance for Engineers

Technical Skills

Programming Languages: Python, R, Julia, Java, VBA, SQL, CoLab
Data Analysis & Statistical Tools: MATLAB, Minitab, Excel Solver, Queuing Theory, JMP, SAS, SPSS
Business Intelligence & Visualization: Power BI, Tableau, Qlik, Google Data Studio, Microsoft Power Platform
Office & Collaboration Tools: Microsoft Excel, PowerPoint, Word, Microsoft 365, Google Workspace, Dynamics 365
Simulation & Engineering Tools: Simio, FlexSim, Gurobi, SCADA, NI Multisim, LabVIEW, AnyLogic, Arena
Process Improvement & Quality Management: AutoCAD, SolidWorks, Root Cause Analysis, Lean Six Sigma, 5 Whys, Control Charts (SPC), Cycle Time Analysis, Value Stream Mapping, SIPOC, Lean Manufacturing, Kaizen

Professional Experience

Sodexo, USA | *Student Manager* *Aug 2025 – Present*

- Promoted to Student Manager; analyzed sales and inventory data using **Excel** to optimize staffing and reduce stock discrepancies, improving operational efficiency
- Utilized data-driven staffing insights to optimize employee scheduling, task allocation, and peak-hour coverage while regularly updating SOPs to improve service efficiency and minimize operational delays

Sula Vineyards | *Harvest Intern* *Jan 2025 – Apr 2025*

- Optimized **30 tons/day** harvest operations using a **multi-sensor** process monitoring approach including load-cell weighing systems, tank-level sensors, pressure and temperature sensors, flow meters, and batch traceability logs
- Created **SOP-based data acquisition** and process-flow records across 10+ production checkpoints, improving intake tracking, fermentation monitoring, transfer control, and scheduling efficiency by 15–20%

Delta Finochem | *Quality Control Intern* *Aug 2023 – Dec 2023*

- Collaborated with **EHS, QA/QC, Operations, and R&D** teams to implement quality and safety standards across API manufacturing, applying **FMEA** and **reliability engineering** principles to improve process reliability and coordination
- Debugged **SCADA**-based ETP control loops for pH/ORP analyzers, ultrasonic level transmitters, electromagnetic flow meters, VFD dosing pumps, aeration blowers, and clarifier units

Project Experience

Discrete-Event Simulation of Ceramic Manufacturing System *Spring 2026*
(Simio / FlexSim)

- Evaluated multiple operational scenarios including capacity expansion, process acceleration, and pull based flow control by developing a discrete-event simulation model in **Simio** and **FlexSim** to analyze a six-stage ceramic manufacturing process and optimize system throughput and resource utilization
- Identified the primary bottleneck and implemented optimization strategies that increased production by 157% (from 7 to 18 units) and reduced waiting time by 86%, improving throughput, reducing work-in-progress inventory, and enhancing overall operational efficiency

Urban EV Charging Demand Forecasting and Infrastructure Optimization *Spring 2026*

- Forecasted urban EV charging demand using Python, MATLAB, and Excel Solver on 10,000+ data points from DOE datasets, improving prediction accuracy by 20% and supporting capacity planning decisions

- Performed data-driven scenario analysis using Minitab and Power BI, optimizing charging infrastructure and reducing projected wait times by 15%

Design Thinking–Driven Development of a Student Networking and Navigation Platform *Fall 2025*

- Applied the Design Thinking **(DEDIPER)** framework to develop Campus Connect, conducting 20+ user interviews and **AEIOU** observations; utilized **Miro** and **FigJam** for empathy mapping, persona creation, and stakeholder analysis to identify key gaps in campus engagement
- Designed and prototyped solutions using **Figma**, creating 10+ wireframes and interactive UI flows; evaluated concepts through prioritization matrices and feedback grids, improving feature selection efficiency by **30%** based on user validation

Isolation and Characterization of Exosome-like Nanoparticles from *Polianthes tuberosa* *Spring 2024*

- Performed process data analysis and visualization using **Python (Pandas, Matplotlib)** and **Excel**, enabling data-driven evaluation of experimental performance and improving workflow repeatability by 15%
- Applied data-driven analysis using **statistical modeling techniques** to monitor experimental workflows, optimize process parameters across 50+ trials, and enhance operational consistency and performance reliability

Leadership Experience

Yuva Marathi (Club at VIT) | *Senior Core Committee* *Aug 2022 – Jan 2024*

- Coordinated club initiatives by supporting event planning, scheduling, and execution in collaboration with the core committee
- Facilitated communication among team members to ensure smooth coordination and timely completion of activities

Biology Learning Center | *Lead Volunteer* *May 2020*

- Led and supported volunteers during educational activities by assigning responsibilities, guiding participants, and maintaining organized workflows
- Ensured smooth operations by assisting instructors and addressing participant needs throughout learning sessions

Publication

Jakhadi, A. P. (2024). *Investigating business strategies for the biotechnology, medical devices, and healthcare sectors to manage uncertainty.* ***International Journal of Scientific Research and Engineering Trends***. ISSN: 2395-566X.

Certification

Lean Six Sigma Green Belt | *Binghamton University* *Expected May 2026*
Lean Six Sigma White Belt | *Binghamton University* *Feb 2026*
Engineering Design Thinking | *Binghamton University* *Jan 2026*
NPTEL Certification | *Indian Institute of Technology* *Apr 2024*
Red Bull | *On-Premise Sales Expert* *Jun 2023*
VIT, Vellore | *Hackathon Participant* *Jun 2023*
Maharashtra State Certificate in Information Technology *Oct 2019*